

PARTICLE IN CELL SIMULATION OF PLASMA THRUSTERS*

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Abstract

A new 3 dimensional Particle-in-Cell code is applied to simulate the operation of a 100 W CHT thruster. In the simulation a spoke moving with a velocity of about 1 km/s was observed. The initial position of the spoke was found to be strongly correlated with the cathode placement. The simulation has shown that the depletion of neutral gas can lead to azimuthal asymmetry of the discharge and possibly to the spoke phenomenon. The spoke and non-spoke regimes of the CHT operation were achieved in the simulation. The simulations have demonstrated that the presence of the spoke visibly changes the ion flow pattern in the thruster.

I. INTRODUCTION

The Hall thruster (HT) [1] is an electromagnetic propulsion device that uses a cross-field plasma discharge to accelerate ions. The thrust is a reaction force to this acceleration, exerted upon the thruster magnetic circuit. In a conventional HT, axial electric and radial magnetic fields are applied in an annular channel. The magnetic field is large enough to lock the electrons in the azimuthal $E \times B$ drift, but small enough to leave the ion trajectories almost unaffected. Because of the reduced electron mobility across the magnetic field, a substantial axial electric field can be maintained in the quasi-neutral plasma and electrons can effectively ionize neutral atoms of the propellant gas. Under such conditions, the electric field supplies energy mainly to accelerate the unmagnetized ions. Unlike the space-charge limited gridded ion engine, the HT accelerates the ions in the quasi-neutral plasma. Thus, larger ion current and thrust densities can be achieved.

The drawback of the annular-geometry of a conventional HT is that it has an unfavorable ratio of the channel surface area to the channel volume. The plasma therefore tends to interact with the thruster channel walls, which results in heating and erosion of parts of the thruster [2]. This tendency becomes more pronounced when the HT is scaled down to low power [3].

A cylindrical geometry Hall thruster (CHT) [4], which was proposed and developed at the PPPL has a reduced

ratio of the channel surface area to volume which is limiting electron transport and ion losses [5,6]. Initially developed for a kilowatt power level [4], this non-conventional HT concept was extensively studied and developed for micro propulsion applications [5-7]. The CHTs demonstrated performance comparable to the state-of-the-art annular HTs of similar power levels [4,8].

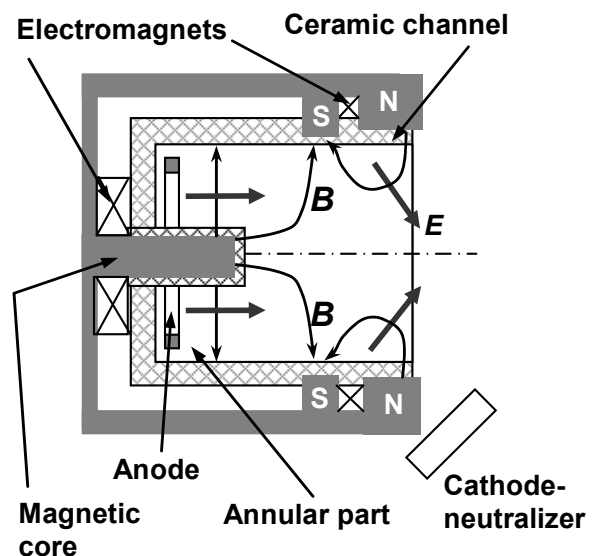


Figure 1. Schematic of a cylindrical Hall thruster [5].

In our previous work [9] we applied a cylindrically symmetrical 2-dimensional Particle-in-Cell code with Monte Carlo Collisions (PIC MCC) [10] to simulate the operation of the PPPL 100 W cylindrical Hall thruster (CHT) [4,7]. Although, the simulation results were in qualitative agreement with probe measurements of potential, electron temperature and plasma density [11], the simulation was not able to reproduce the dependence of the plasma parameters on the enhancement of cathode electron current observed in the experiment [11]. The possible reason for discrepancy is that in the simulation azimuthal dynamics was missing and anomalous electron transport perpendicular to magnetic field was included in

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the simulation only empirically via Bohm-type diffusion with fixed diffusion coefficient.

In order to resolve the electron cross-field transport self-consistently, accounting for plasma azimuthal dynamics, we have developed a new fully three-dimensional PIC MCC code STOIC. The first results of the 3D code for 100 W CHT are presented in this paper.

II. CYLINDRICAL HALL THRUSTER

Fig. 1 illustrates the principle of operation of the cylindrical thruster. A cylindrical Hall thruster consists of a cylindrical ceramic channel, a ring-shaped anode, which serves also as a gas distributor, a magnetic core and magnetized sources (Fig. 1). The magnetic field lines intersect the ceramic channel walls. The electron drifts are closed, with the magnetic field lines forming equipotential surfaces [4]. The radial component of the magnetic field crossed with the azimuthal electron current produces the thrust. However, the electrons are not confined to an axial position; rather they bounce over an axial region, impeded from entering the annular part of the channel because of magnetic mirroring. Two magnetized sources, electromagnetic coils with opposite currents, can produce a cusp-like magnetic field in the channel, with a strong radial component. To maintain ionizing collisions, the anode (gas inlet) is placed in the short annular part of the channel. The length of the annular part of the channel is designed to minimize the ionization mean free path, thus localizing the ionization of the working gas at the boundary of the annular and cylindrical regions. Hence, most of the voltage drop should occur in the cylindrical region that has large volume-to-surface ratio. This conclusion is supported by the results of plasma measurements [4,7] and by Monte-Carlo simulations [6].

III. THE MODEL

The detailed description of the PIC-MCC method can be found in thorough reviews [12,13]. Here, we just outline the specific features of our model. In PIC-MCC simulations we follow the kinetics of so-called “Super Particles” (each of them representing many real particles), moving in the self consistent electric field calculated on a spatial grid from the Poisson equation. The particle collisions are handled by Monte-Carlo collision (MCC) routines, which randomly change particle velocities according to the actual collision dynamics. The simulation includes electrons, Xe^+ ions and the neutral Xenon atoms. All relevant collisional processes are included in the model: electron-neutral elastic, ionization and excitation collisions, ion-neutral momentum- transfer and charge exchange collisions. The dynamics of the background neutral gas is self-consistently resolved with direct simulation Monte Carlo.

In our model we resolve 3 spatial and 3 velocity components. The model utilizes an equidistant Cartesian grid which explicitly assures momentum conservation and zero self forces. The computational domain in the present simulations is extended beyond the discharge channel and includes the near-field region of the thruster.

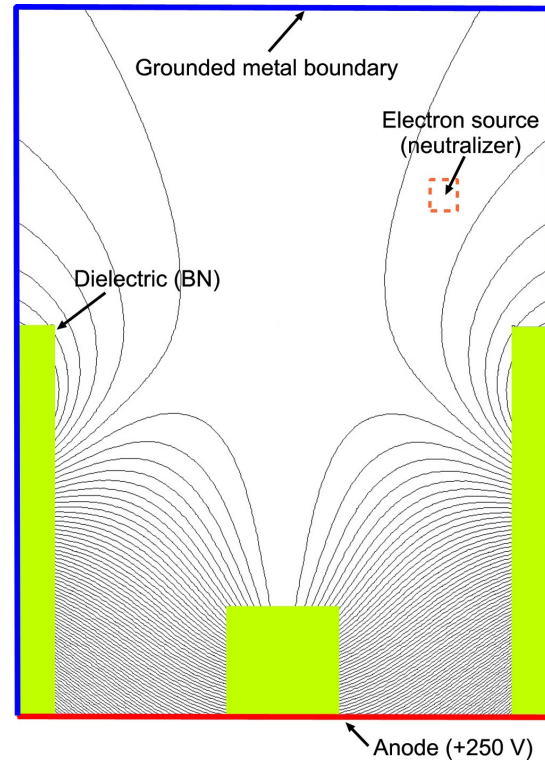


Figure 2. Computational domain with magnetic field topology.

The computational domain represents a parallelepiped with length $Z_{\max} = 40$ mm and sides $X_{\max} = Y_{\max} = 30$ mm. Z axis is directed along the thruster symmetry axis. The axial cross-section of the computational domain together with magnetic field topology is shown in Fig. 2. The thruster channel length (distance from the anode to the exit plane) is $Z_{thr} = 22$ mm. The thruster channel walls are located at $X_{ch1} = 3$ mm, $X_{ch2} = 27$ mm, $Y_{ch3} = 3$ mm, $Y_{ch4} = 27$ mm. At $Z = 0$ the metal anode is located. At the center of the anode the annular part with the length $Z_{anular} = 6$ mm and sides $X_{anular} = Y_{anular} = 6$ mm is placed. The region outside the thruster exit: ($Z_{thr} < Z < Z_{\max}$, $X_{ch1} < X < X_{ch2}$, $Y_{ch1} < Y < Y_{ch2}$) represents the very near-field plume zone. The thruster channel and the annular part are dielectric (Boron Nitride). The lateral and axial boundaries of the computational domain are assumed to be metallic. All metal elements in the

simulation, except for the anode are at ground potential. At the anode, a voltage of $U_a = 250$ V is applied.

In order to compute the electrostatic potential in the system, the Poisson equation is discretized on the grid, taking into account the possible change of the dielectric permittivity ε across the dielectric surfaces. The set of resulting finite difference equations is solved with lower-upper factorization technique using Watson Sparse Matrix Package [14]. This approach allows us to calculate the potential inside the computational domain, self-consistently resolving the floating potential on the dielectric surfaces.

The particles' equations of motion are solved for discrete time steps with the leap-frog / Boris algorithm [12]. All surfaces in the simulation are assumed to be absorbing for electrons and ions. No secondary electron emission is assumed in the simulation. The neutrals are re-launched from the surfaces using a Maxwellian distribution with temperature $T_n = 1000$ K. The neutral gas atoms are injected into the system through two coaxial ring slits at the anode with the mass flow rate $\dot{m} = 0.4$ mg/s.

Electrons with a Maxwellian distribution and a temperature $T_e = 5$ eV are introduced into the system in the source region $51 \text{ mm} < X < 53 \text{ mm}$, $13 \text{ mm} < Y < 17 \text{ mm}$, $26 \text{ mm} < Z < 27 \text{ mm}$ with uniform density and the constant current $I_c = 0.25$ A, simulating the thruster cathode. In the simulation the electrons from the source, accelerated in the thruster's electric field, are ionizing the neutral gas, creating the plasma in the thruster channel. In order to speed up the plasma ignition process, the supplementary ring source of the electrons with temperature $T_e = 20$ eV and current $I_c = 0.05$ A was applied in the annular part during the first 300 ns of the simulation. After that the supplementary source was switched off and only the cathode electron source was operating.

To reduce the computational time the size of the system is scaled down by factor of 10. In order to preserve the ratio of the particles mean free paths and the gyroradii to the system length, the collisions cross-sections and the magnetic field are increased by the same factor 10.

An equidistant computational grid $60 \times 60 \times 80$ was used in the simulation. The total number of computational particles in the simulation was about 40000000. The cell size $\Delta x = \Delta y = \Delta z = 5 \cdot 10^{-2}$ mm in the simulation was chosen to ensure that it is smaller than the smallest Debye length in the system. The time step was set to $\Delta t = 5.6 \cdot 10^{-12}$ s in order to resolve the electron plasma frequency. The simulation was carried on a 4-processor Intel Xeon workstation. The duration of the run was about 20 days. About $4 \cdot 10^6$ time steps were performed which corresponds to a simulated time of 24 μ s.

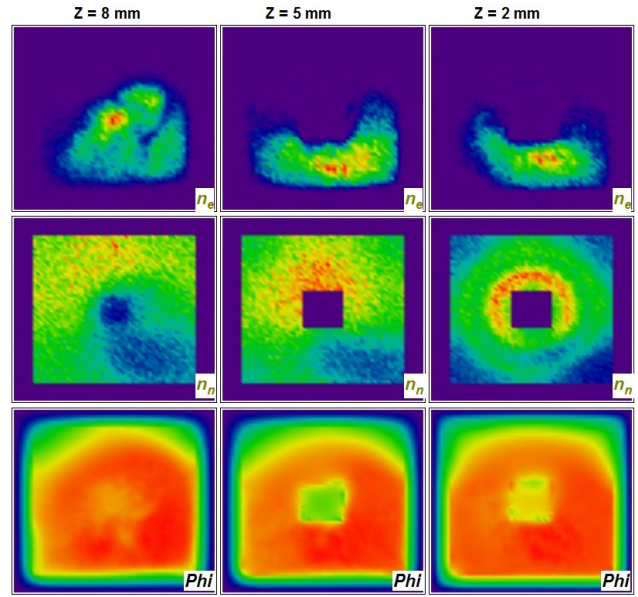


Figure 3. Snapshot of the plasma parameters at $t = 1.7 \cdot 10^{-6}$ s taken at six lateral cross-sections. Top row - plasma density, middle row - neutral density, bottom row - plasma potential. All values are given in relative units: reds correspond to the maximum value, blues to the minimum.

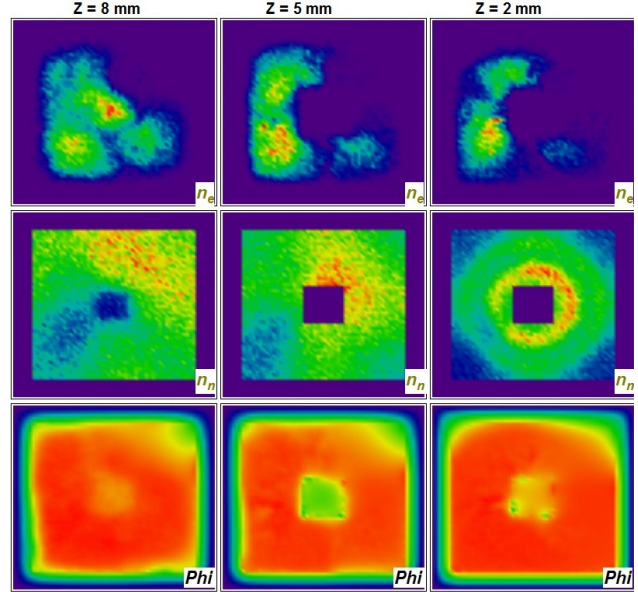


Figure 4. Snapshot of the plasma parameters at $t = 2.9 \cdot 10^{-6}$ s taken at six lateral cross-sections. Top row - plasma density, middle row - neutral density, bottom row - plasma potential.

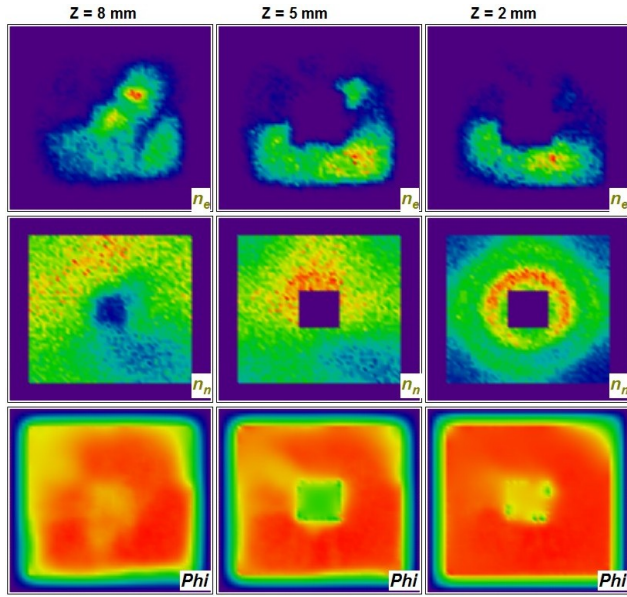


Figure 5. Snapshot of the plasma parameters at $t = 4.3 \cdot 10^{-6}$ s taken at six lateral cross-sections. Top row - plasma density, middle row - neutral density, bottom row - plasma potential.

The results of the simulations are summarized in Figs. 3-5. At each of the figures snapshots of the electron density, the neutral density and the plasma potential, taken at lateral cross-sections at three axial positions, are presented. The $Z = 2$ mm and $Z = 5$ mm correspond to the annular part of the thruster channel, $Z = 8$ mm is in the cylindrical part of the thruster channel.

In Fig. 3 one can see the parameters at $t = 1.6 \cdot 10^{-6}$ s of the simulation. By this time a pronounced axial asymmetry in the electron density is formed in the thruster channel. In the annular channel a well localized plasma “spoke” appears. Rotating spokes in the plasma density were observed in numerous Hall thruster experiments [15-21].

The initial position of the spoke was found to be strongly correlated with the cathode placement. In the simulation with same parameters but with a cathode azimuthal position swapped onto the other side (rotated by 180°), the spoke changed its position approximately by 180° . Thus, in our simulation the asymmetry caused by the cathode plays an important role for the origin of the spoke. In the experiment other factors can contribute to the asymmetry as well - neutral gas injection, partial contamination of the anode, asymmetry in magnetic field, etc. In Fig. 3 it is clearly visible that the spoke causes azimuthal asymmetry in the neutral density profile along the thruster channel, depleting the neutrals due to electron-impact ionization. The depletion of the neutral density at the spoke position reaches $\sim 80\%$ at $Z = 8$ mm. This is the region where the electron density reaches its

maximum $n_{e \max} = 2.4 \cdot 10^{12} \text{ cm}^{-3}$ along the thruster channel. The rapid ionization in the spoke indicates that a strong electric field is present, heating the electrons. Indeed, the electric field oscillations with frequency $f \approx 10 \text{ MHz}$, wavelength $\lambda \approx 2\text{-}4 \text{ mm}$ and the amplitude up to 100 V/cm both in the axial and azimuthal directions are observed in the simulation. These fluctuations are responsible for the electron heating and cross-field transport in the simulation. The electron current through the spoke in the simulation was $I_e \approx 0.55 \text{ A}$, which is at least by two orders of magnitude larger than current provided by “classic” diffusion due to collisions with neutrals; and no near-wall conductivity was accounted in the model, as the secondary electron emission was neglected.

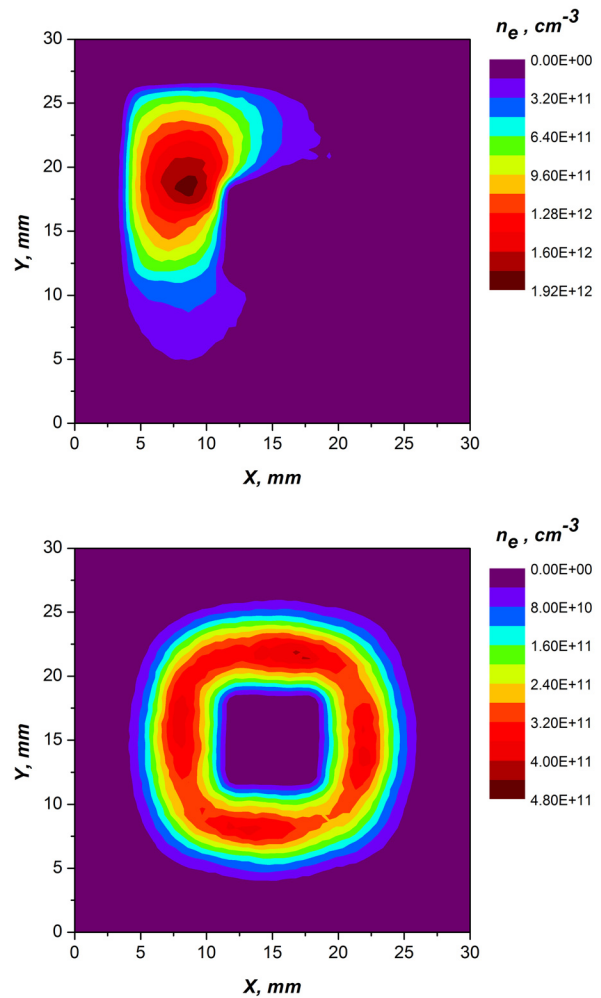


Figure 6. Electron density in the annular channel of CHT thruster: cathode current $I_{cath} = 0.25 \text{ A}$ - spoke regime (top); $I_{cath} = 0.5 \text{ A}$ - non-spoke regime (bottom).

In Figs. 3-5 one can see how the spoke changes its azimuthal position towards the region where the neutral

density is not depleted. Within $\sim 1.2 \mu\text{s}$, the spoke moves, rotating about 90° in the direction of $\mathbf{E} \times \mathbf{B}$ drift. The average velocity of the spoke is about 0.8 km/s , which is a factor ~ 2 lower than was observed in the experiments [19–21]. By the time $2.9 \mu\text{s}$ the initial depletion of the neutral density (bottom right corner) is completely refilled whereas the new depleted region is created at the current spoke position (bottom left corner). At this time the spoke starts to change its position back to its previous place, but instead of rotating the new spoke is originated in the bottom left corner and the first one is dissolved. If one interprets this dynamics as directed motion the corresponding velocity is again about 1 km/s .

The spoke and non-spoke regimes of the CHT operation were achieved in the simulations. In agreement with the experiment [11], the transition from spoke to non-spoke regime in the simulations was observed with an increase of the electron current from the cathode. In Fig. 6 the electron density profile inside the annular channel for cathode currents $I_{\text{cath}} = 0.25 \text{ A}$ and $I_{\text{cath}} = 0.5 \text{ A}$ are shown.

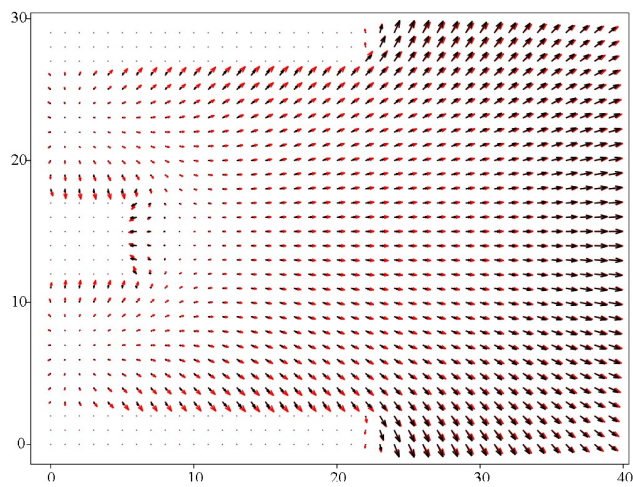


Figure 7. Averaged ion flow map for spoke (black) and non-spoke regimes (red).

In order to investigate the effect of the spoke on the ion flow in the thruster, the ion flow velocity was averaged over 5 cycles of the spoke rotation. The averaged ion velocity vector map at the longitudinal axial cross-section of the simulation domain for spoke and non-spoke regimes is presented in Fig. 7. As one can see from the simulation results, in the spoke regime ion velocity in the thruster channel is distinctly lower as compared with non-spoke regime. Furthermore, the simulated ion flow velocity in the spoke regime outside of the thruster, away from centerline, exhibits visibly higher divergence as compared with non-spoke cathode overrun regime. This feature agrees well with experimental measurements in [22]. In the spoke regime the largest part of the ion

acceleration takes place outside of the thruster, in the plume. In contrast, without the spoke the largest part of acceleration happens inside the thruster channel. The maximum exhaust ion axial velocity in the simulations in both cases is about 16 km/s . The axial ion velocity at the thruster exit is 5.4 km/s with the spoke and 9.2 km/s without spoke. This results agree quite good with findings in [22].

Although the mechanism of the spoke formation and dynamics as well as the anomalous transport inside the spoke is still to be understood, in our simulation we have observed that the spoke motion is associated with the strong azimuthal depletion of the neutral gas, which indicates the importance of the ionization for this instability. The clarification of the phenomena underlying the spoke formation and the dynamics will be the goal of our further research.

IV. SUMMARY

A new, fully three-dimensional PIC MCC code STOIC was developed. The code was applied to simulate the operation of a 100 W CHT thruster. In the simulation a spoke moving with the velocity of about 1 km/s was observed, which is a factor ~ 2 lower than in the experiments. The initial position of the spoke was found to be strongly correlated with the cathode placement. The simulation has shown that the depletion of neutral gas leads to azimuthal asymmetry of the discharge and to the spoke phenomenon.

The simulations have demonstrated that the presence of the spoke visibly changes the ion flow pattern in the thruster. With the spoke the ion flow divergence outside of the thruster is higher as compared with non-spoke cathode overrun regime. In the spoke regime the largest part of the ion acceleration takes place outside of the thruster, in the plume. In contrast, without the spoke the largest part of acceleration happens inside the thruster channel.

The clarification of the phenomena underlying the spoke formation and the dynamics as well as electron transport inside the spoke will be the goal of our further research.

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